



TailorSync

User Manual

AI-Powered Tailoring Business Management System

Application	TailorSync (Android app & web app)
Application version	1.0.0
Group number	Group 07
TCC module	CCS2360 – Technology Challenge Competition
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Faculty	Faculty of Computing & IT, Sri Lanka Technology Campus

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1. System Overview

TailorSync helps small and medium tailoring shops manage customers, measurements, orders, staff tasks and fabric stock in one mobile app, and uses AI to predict measurements, recommend fabric and preview garments on customers.

Target users: shop **Owners** (full access) and **Staff** such as tailors and cutters (see and update their own tasks).

Main features: customer records · 9-step AI Order Wizard · 7-stage production tracking · cutting blueprints · fabric inventory · reports & charts · staff management.

AI/ML functionality

- Measurement prediction** – Custom ML model (shirts & trousers) or Azure AI Foundry (all garments).
- Fabric recommendation & estimation** – best fabrics with a match % and metres required.
- Virtual Try-On** – AI image of the customer wearing a chosen garment.

2. System Requirements & Access

Item	Requirement
Device / OS	Android 8.0 or later phone/tablet, or a computer with the latest Google Chrome / Microsoft Edge
Internet	Wi-Fi or mobile data is required (login, saving data and all AI features run on the cloud server)
Software	No extra software. Camera (optional) for Virtual Try-On photos
Installation	Android: install the TailorSync APK supplied with this submission (allow “Install unknown apps” when asked) and open TailorSync. Web: open the TailorSync web link in Chrome
Login	Owner: tap Create account and register the business. Staff: use the email and temporary password sent by the owner. Tick “I’m not a robot” , then Sign In

3. How to Use the System

3.1 Sign in and Home dashboard

- Open TailorSync and enter your **Email** and **Password** (forgot it? tap **Forgot Password?**).
- Tick **I’m not a robot** and tap **Sign In**. The **Home** dashboard shows today’s Active, Due Today, Overdue and Completed orders with shortcuts.
- Use the blue bar to move between **Home, Orders, Customers, Tasks** and **Reports**.

3.2 Add a customer

Customers → **Add New** → enter **Full name** and **Phone number** (email and address optional) → **Save Customer**.

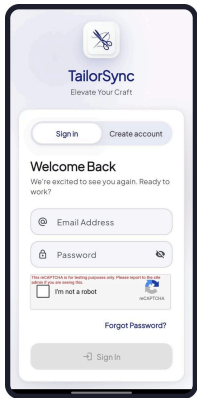


Fig 1. Login with reCAPTCHA

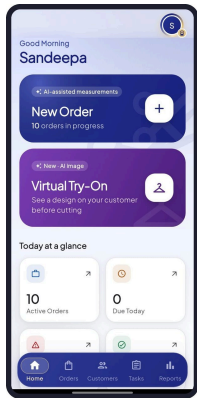


Fig 2. Home dashboard

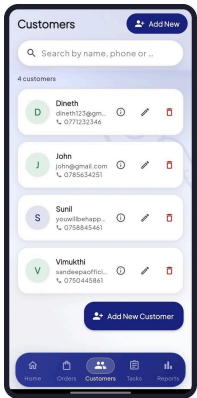


Fig 3. Customers list

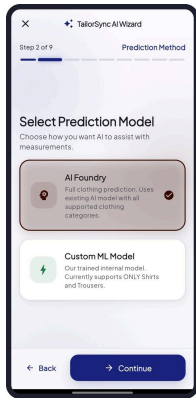


Fig 4. Choose the AI model

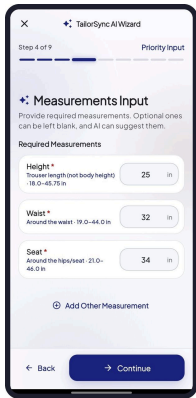
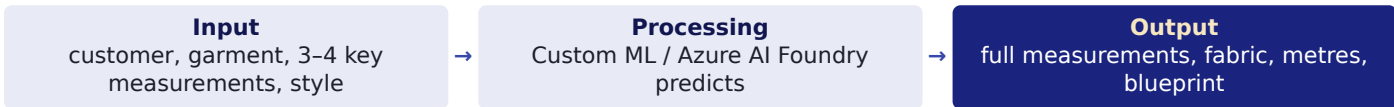


Fig 5. Enter key measurements

3.3 AI/ML workflow - create an order with AI prediction (Input → Processing → Output)



- On Home tap **New Order**. **Step 1:** select the customer.
- Step 2:** choose the AI model – **AI Foundry** (all garments) or **Custom ML Model** (fast; shirts & trousers) (Fig 4).
- Step 3:** choose the garment, e.g. *Short Trouser*. **Step 4 - Input:** type the required measurements in inches, e.g. Height 25, Waist 32, Seat 34. The allowed range is shown under each field (Fig 5). Tap **Continue**.

- Step 5 - Processing & result:** the screen shows “Microsoft Foundry is Thinking...” (Fig 6). The AI then shows **Recommended Measurements** for every missing value (e.g. Round Knee 20.0, Crotch 9.5) and **Alternatives** (Fig 7). Select the best option; tap **Edit Measurements** to change any value.
- Step 6:** select Occasion, Weather and Fit. **Step 7 - Output: Top Fabrics** with a **match %** (e.g. Merino Wool Blend 92%) and stock status (Fig 8). Tap a fabric.
- Step 8 - Output: Fabric Required**, e.g. **0.75 metres**, with a range; type a different amount in Override Quantity if needed (Fig 9).
- Step 9:** check the Order Summary, optionally add a price, choose **Assign to Staff** and tap **Save Order**.

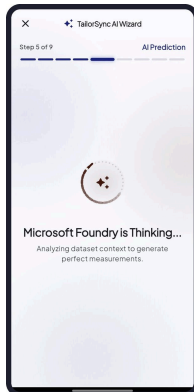


Fig 6. AI processing

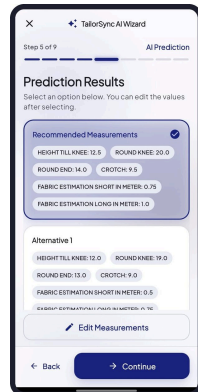


Fig 7. Predicted measurements

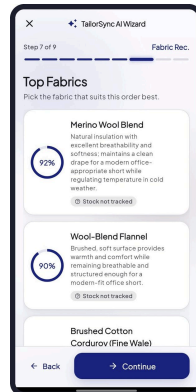


Fig 8. Fabric recommendations



Fig 9. Fabric estimate

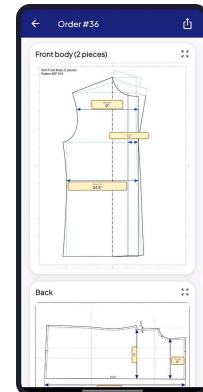


Fig 10. Blueprint with values

What the result means: the predicted measurements complete the customer’s size from only a few inputs, so fewer measurements are needed. The match % shows how suitable each fabric is for the garment, occasion and weather, and the metre estimate tells you how much fabric to cut or buy. Saved values appear in **Order details** and on the **cutting blueprint** (Fig 10). AI results are suggestions – always check them before cutting.

3.4 Track production

Tasks → on a task tap **Update Stage** → choose the next stage (Order Received → Cutting → Sewing → Fitting → Quality Check → Ready → Delivered) → **Update Stage**. Tap the due date to change it.

3.5 Virtual Try-On (AI image)

Home → **Virtual Try-On** → add a **Camera/Gallery** photo → choose **Garment** (e.g. Kandyan saree), **Colour** and **Fabric** → tick the consent box → **Generate Try-On**. The output is a picture of the customer wearing that outfit, to help them choose before cutting.

3.6 Fabric Inventory and Reports (Owner)

Inventory shows metres in stock; saved orders deduct fabric automatically and low stock is flagged. **Reports** shows orders, on-time delivery, fabric used and AI usage for 7/30/90 days.

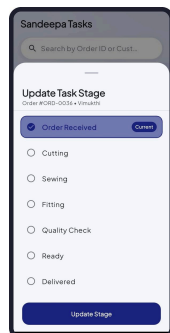


Fig 11. Update task stage

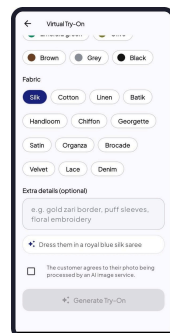


Fig 12. Virtual Try-On options

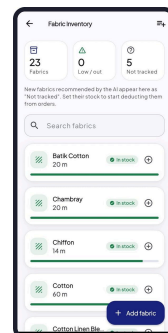


Fig 13. Fabric Inventory

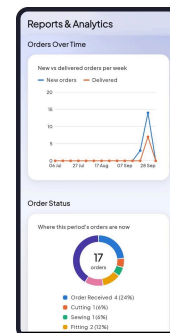


Fig 14. Reports & charts

4. Troubleshooting

Problem	Solution
Cannot sign in / Sign In button grey	Check email and password, fill both fields and tick “I’m not a robot”. Use Forgot Password? if needed.
“Required” or out-of-range measurement	Complete all fields marked * and enter values in inches inside the range shown under the field.
Custom ML not available / AI is slow	Custom ML supports only shirts and trousers – choose AI Foundry. Check the internet connection and tap Continue again.
“Today’s free image limit is used up”	The daily Virtual Try-On limit was reached; try again after 5:30 AM (Sri Lanka time).